VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.

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Best of luck!

PAST PAPER

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

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Answer (Please select your correct option)

- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\text{--------}\frac{5}{6}\frac{3}{4}\frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\text{--------}\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\text{--------}\frac{6}{7}\frac{4}{5}\frac{2}{3}$ ⇦
- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\text{--------}\frac{6}{5}\frac{4}{3}\frac{2}{1}$

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4 \sin 2x$?

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Answer (Please select your correct option)

- 2
- 4
- 8
- 16

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

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Answer (Please select your correct option)

- $\frac{\pi}{2}$
- π
- $\frac{3\pi}{2}$
- 4π

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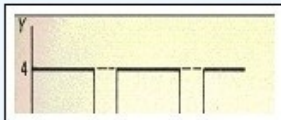
Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

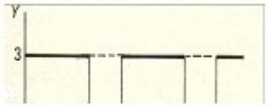
Match the following periodic function with its graph.

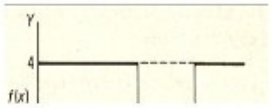
$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

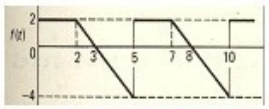
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Answer (Please select your correct option)









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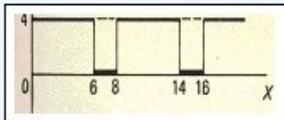
Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

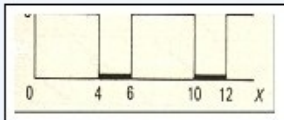
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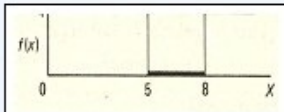
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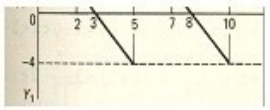
Answer (Please select your correct option)





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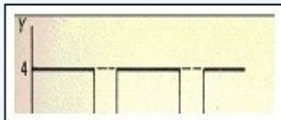
Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

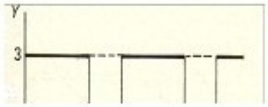
Match the following periodic function with its graph.

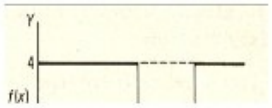
$$f(x) = \begin{cases} 4 & 0 < x < 6 \\ 0 & 6 < x < 8 \end{cases}$$

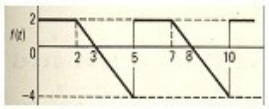
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Answer (Please select your correct option)









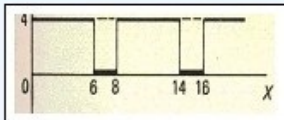
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Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Match the following periodic function with its graph.

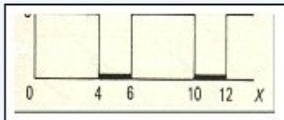
$$f(x) = \begin{cases} 4 & 0 < x < 6 \\ 0 & 6 < x < 8 \end{cases}$$

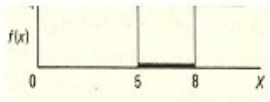
Answer (Please select your correct option)

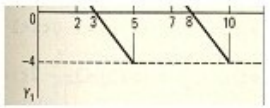


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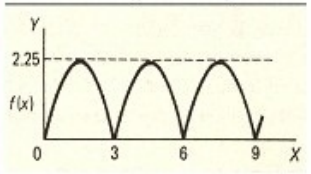


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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



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Answer (Please select your correct option)

- 0
- 2.25
- 3 correct
- 6

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

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Answer (Please select your correct option)

- Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken.
- Divergence of $\vec{F}$ should be zero
- Gradient of $\vec{F}$ should be zero.
- $\vec{F} = 0$

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Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Let L denotes the Laplace Transform.
According to First Shift Theorem, if $L\{F(t)\} = f(s)$ then which of the following equation holds?
s *and* a are constants.

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Answer (Please select your correct option)

- $L\{e^{-at}F(t)\} = f(s-a)$
- $L\{e^{-at}F(t)\} = f(s+a)$
- $L\{e^{-at}F(t)\} = f(s)$
- $L\{e^{-at}F(t)\} = f(a)$

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 \cos 2x$ is ----------

Answer (Please select your correct option)

- Even function
- Odd function
- Neither even nor odd

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

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Answer (Please select your correct option)

- $3te^{2t} - 2$
- $3e^{2t} - 2t$
- $3e^{2t} - 2$ correct
- None of these.

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace transform of a function F(t)?

(s is a constant)

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Answer (Please select your correct option)

- $\int_0^s e^{-st} F(t)\, dt$
- $\int_0^\infty e^{st} F(t)\, dt$
- $\int_{-\infty}^\infty e^{-st} F(t)\, dt$
- $\int_0^\infty e^{-st} F(t)\, dt$ correct

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

Answer (Please select your correct option)

- $L\{\sin 3t\} = \frac{3}{s^2+9}$ correct
- $L\{\sin 3t\} = \frac{s}{s^2+9}$
- $L\{\sin 3t\} = \frac{1}{s-3}$
- $L\{\sin 3t\} = \frac{3!}{s^4}$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If L denotes laplace transform then
$L(te^{5t})$ =

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Answer (Please select your correct option)

- $L(te^{5t}) = \frac{1}{s^2-5}$
- $L(te^{5t}) = \frac{1}{s^2+5}$
- $L(te^{5t}) = \frac{1}{(s+5)^2}$
- $L(te^{5t}) = \frac{1}{(s-5)^2}$ correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{s}{s^2+25}$

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Answer (Please select your correct option)

- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \sin 5t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 5t$ correct
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 25t$

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

What is $L(-6)$ if L denotes Laplace Transform?

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Answer (Please select your correct option)

- $L\{-6\} = \frac{1}{s+6}$
- $L\{-6\} = \frac{-6}{s}$ correct
- $L\{-6\} = \frac{s}{s^2+36}$
- $L\{-6\} = \frac{-6}{s^2+36}$

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Curl of vector function is always a ----------

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Answer (Please select your correct option)

- Scalar
- Vector

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is geometrical representation of set of real numbers?

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Answer (Please select your correct option)

- Co-ordinate line
- xy-plane
- Sphere
- Circular cylinder

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the interval notation of real line?

Answer (Please select your correct option)

- (-∞ , +∞)
- (-∞ , 0)
- (0 , +∞)

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

An ordered triple corresponds to ---------- in a three dimensional space.

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Answer (Please select your correct option)

- A unique point
- A point in each octant
- Three points
- Infinite number of points

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the distance between points (3, 2, 0) and (1, 0, -1)?

Answer (Please select your correct option)

- 3 correct
- $\sqrt{6}$
- $\sqrt{3}$
- $\sqrt{10}$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

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Answer (Please select your correct option)

- 3, 2, 9
- 1, -4, -1
- 2, -3, 20
- 0.5, -3, 5/4

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

In three dimensional space, the equation $y = x^2$ always represents . . . . . .

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Answer (Please select your correct option)

- Parabola

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- Straight line
- Half cylinder
- Cone

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x,y,z)$, the equation $\frac{\partial^2 f}{\partial^2 x}+\frac{\partial^2 f}{\partial^2 y}+\frac{\partial^2 f}{\partial^2 z}=0$ is known as ----------

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Answer (Please select your correct option)

- Gauss Equation
- Euler's equation
- Laplace's Equation
- Stoke's Equation

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Made by: Waqar Siddhu

Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Every differentiable function is always . . . . . .

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Answer (Please select your correct option)

- Piece wise continuous
- Discontinuous
- Continuous

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a . . . . . function.

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Answer (Please select your correct option)

- Scalar
- Continuous
- Vector
- Constant

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

The two normal lines for the surfaces $f(x,y,z) = 0$ and $g(x,y,z) = 0$, are orthogonal if and only if . . . . . where f_x, f_y, f_z and g_x, g_y, g_z are direction ratios for the two normal lines respectively.

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Answer (Please select your correct option)

- $f_x g_y + f_y g_z + f_z g_x = 0$
- $f_x g_x + f_y g_y + f_z g_z \geq 0$
- $f_x g_x + f_y g_y + f_z g_z = 0$ correct
- $f_x + f_y + f_z + g_x + g_y + g_z = 0$

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

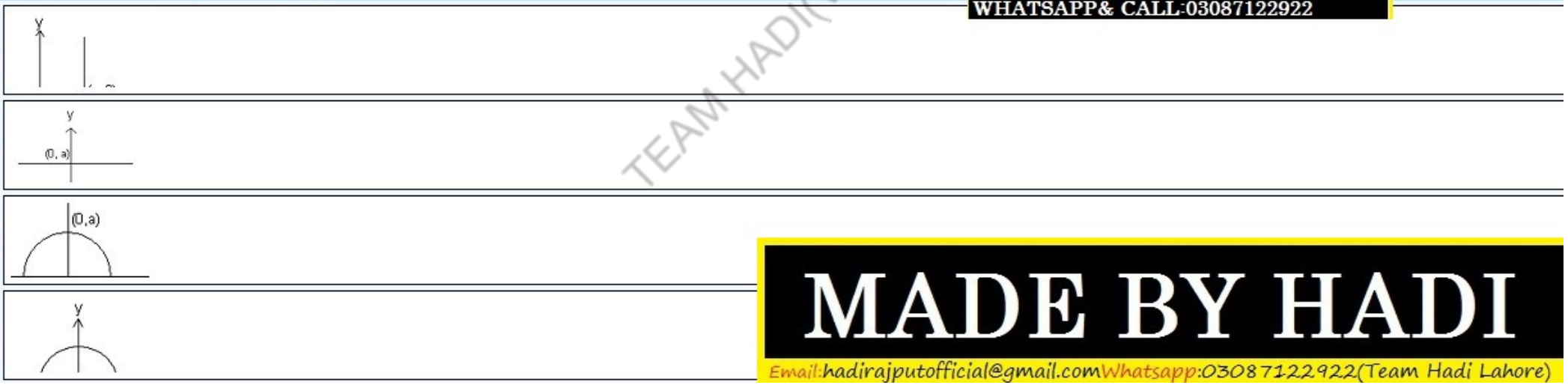
Match the following equation in polar co-ordinates with its graph.

$r \cos\theta = a$

where a is an arbitrary constant

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Answer (Please select your correct option)



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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

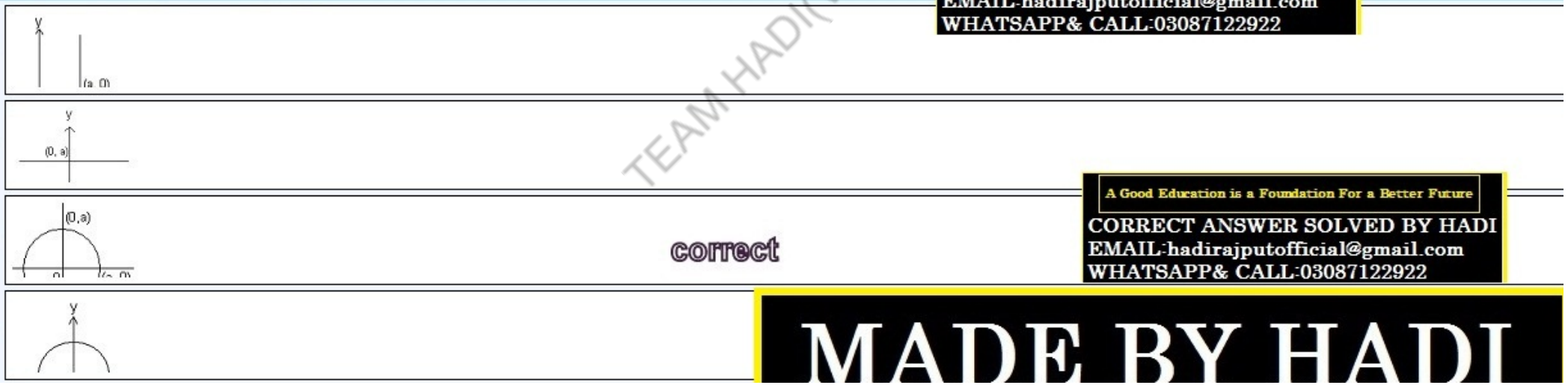
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where a is an arbitrary constant

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Answer (Please select your correct option)



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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

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Answer (Please select your correct option)

- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$ correct
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

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Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

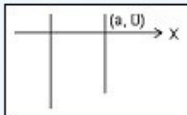
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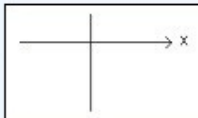
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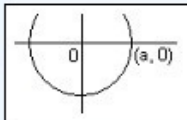
where a is an arbitrary constant

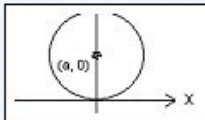
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Answer (Please select your correct option)









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Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

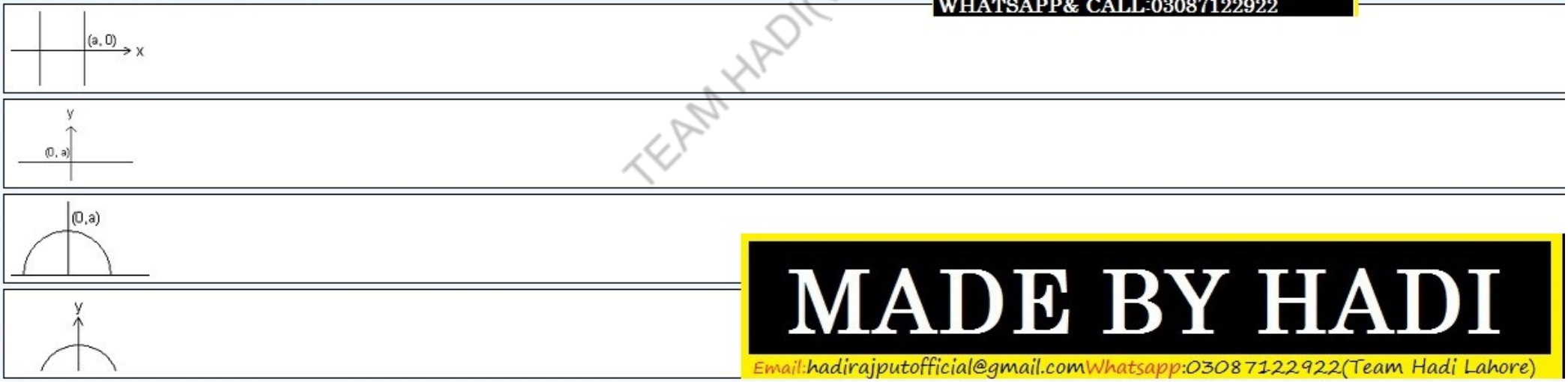
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Answer (Please select your correct option)



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Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

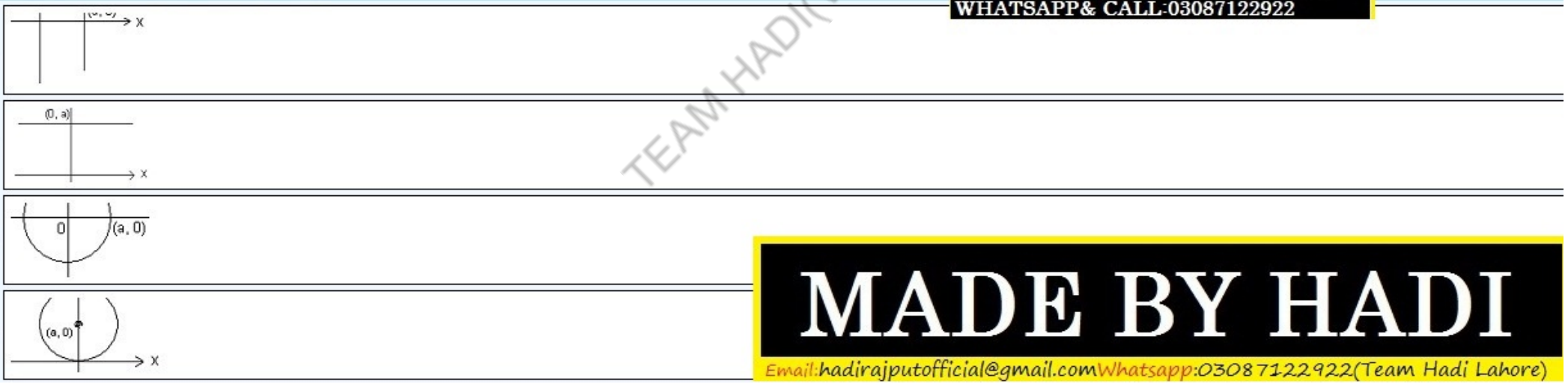
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Answer (Please select your correct option)



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Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

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Answer (Please select your correct option)

- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, -\theta)$ then the curve is said to be symmetric about .....

Answer (Please select your correct option)

- Initial line
- y-axis
- Pole
- origin

correct

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . . where $a \le \theta \le b$ and c ≤ r ≤ d.

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Answer (Please select your correct option)

- $\int_a^b \int_c^d f(r,\theta)rdrd\theta$
- $\int_a^b \int_c^d f(r,\theta)drd\theta$ correct
- $\int_a^b \int_c^d f(r,\theta)rd\theta dr$
- $\int_a^b \int_c^d f(r,\theta)d\theta dr$

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

The parametric equations that correspond to the vector equation $\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$ are . . . . . .

Answer (Please select your correct option)

- $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 0$
- $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z = 0$
- $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 1$
- $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of the equation $r = 3\hat{i} - 2\hat{j} - \hat{k}$ is the point . . . . . .

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Answer (Please select your correct option)

- (1, −2, 3)
- (3, −2, −1)
- (3, 2, 1)
- (−3, −2, −1)

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\vec{R}(t)$ is the anti-derivative of a given vector valued function $\vec{r}(t)$ i.e., $\vec{R}'(t) = \vec{r}(t)$ then $\int_a^b \vec{r}(t)dt = \ldots\ldots\ldots$

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Answer (Please select your correct option)

- $\vec{R}(a) + \vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a) + c$ where c is a non-zero constant.
- $\vec{R}(a).\vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a)$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t)\times\vec{r}_2(t)]$ =..... where "×" and "▪" represent the cross and dot product respectively.

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Answer (Please select your correct option)

- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \vec{r}_2 \times \frac{d\vec{r}_1}{dt}$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector valued function $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is continuous if . . . . . .

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Answer (Please select your correct option)

- Atleast one of its components is continuous.
- All of its components are necessarily differentiable.
- All of its components are continuous.
- Limit exists for all of its components.

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Question No : 37 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Given a vector valued function $\vec{r}(t) = \frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ and its anti-derivative $\vec{R}(t) = \ln(t-3)\hat{i} + e^t\hat{j}$, then $\int \vec{r}(t)dt = \ldots\ldots\ldots\ldots$

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Answer (Please select your correct option)

- $\ln(t-3)\hat{i} + e^t\hat{j} + c$ correct
- $(t-3)\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $(t-3)^{-1}\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $\frac{1}{(t-3)}\hat{i} + e^t\hat{j}$

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t).\vec{r}_2(t)] =$ ..... where "×" and "." represent the cross and dot product respectively.

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Answer (Please select your correct option)

- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \times \vec{r}_2$ ⇦

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A single curve can be represented by . . . . . vector valued function(s).

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Answer (Please select your correct option)

- Two
- Infinitely many
- Single
- Three

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

A function is said to be smooth if it's derivative is . . . . . on any value of its domain.

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Answer (Please select your correct option)

- continuous and non zero
- piecewise continuous
- defined and non zero
- differentiable ⇦

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis cosine formula to evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$

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Answer (Please click here to Add Answer)

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^3 + x^2$

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Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Let $f(x, y) = \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$. Is the function defined at (1, 1)? If yes, what is its value and if no, give the reason.

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Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate the following limit.

$$\lim_{t \to \frac{\pi}{4}} \left[(\cos t)\hat{i} + (\sin t)\hat{j} \right]$$

Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the fourier co-efficient a_0, of periodic function defined by

$f(x) = x \qquad 0 < x < 1$

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Answer (Please click here to Add Answer)



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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^4 x\right) dx$

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Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If the order of integration for the integral $\int_0^2 \int_{y^2}^4 y \cos x^2 dx dy$ is changed. Find the change in the limits of new integral.

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Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ when $0 \le t \le 2\pi$?

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Answer (Please click here to Add Answer)

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Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider a periodic function defined by

$f(x) = 3x \qquad -\pi \le x \le \pi$

(i) Find whether the given function is even or odd?

(ii) Determine Fourier Co-efficients a_0, a_n and b_n

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Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\,\hat{j} + xz^2\,\hat{k}$

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Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Find Equation of a Tangent plane to the surface $f(x, y, z) = x^2 + 3y + z^3 - 9$ at the point $(2, -1, 2)$

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Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + (t^2 - 5)\,\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r}'(t)$ are perpendicular to each other.

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Answer (Please click here to Add Answer)

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